

GENESIS

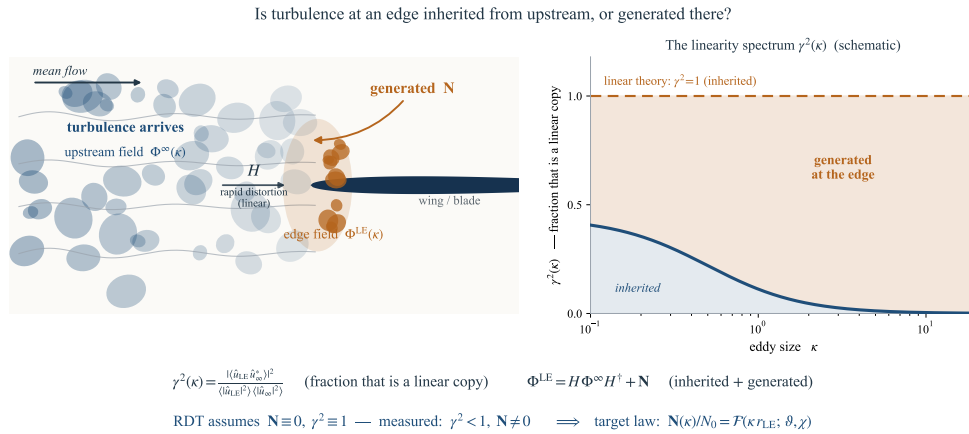
The birth of sound-producing turbulence at a boundary

Principal Investigator: Dr. Sparsh Sharma
Host Institution: German Aerospace Center (DLR), Braunschweig, Germany
Proposal acronym: GENESIS
Panel: PE8
Project duration: 60 months
Call: ERC-2027-StG

Abstract

Whenever turbulent air impinges on a solid body — a wind-turbine blade, a rotor, a fan — a thin, strongly strained layer at the body's edge sets the unsteady loads the machine carries and the noise it radiates. For seventy years this layer has been described through one largely untested assumption: that the turbulence arriving at the surface is a *linear image* of the flow upstream (rapid-distortion theory). A second assumption sits beneath it: that the acoustic *source* is a well-defined object at all. **GENESIS subjects both assumptions to direct test, and develops the two theories required where they fail.** Recent results by the PI call each assumption into question: the first scale-resolved measurement of the linearity of the transformation indicates that the turbulence responsible for the sound is largely *generated at the edge* rather than inherited — through a pressure–strain mechanism outside the scope of linear theory, organised by the edge radius — while an exact *transport geometry* derived by the PI shows that “the source” is gauge-dependent, with a computable structure. From these starting points, GENESIS develops two theories. *First*, a **non-equilibrium theory of strongly strained turbulence**, applicable where linear RDT and equilibrium closures cease to hold, predicting the generated turbulence from geometry and strain history, and testing whether boundary-induced generation constitutes a **new universality class** or a critical crossover — mapping, with measured exponents, whichever it proves to be. *Second*, an **invariant, geometric theory of the aeroacoustic source**: the part of the turbulent field that is holonomy-invariant, and therefore radiates, extended from the constant-coefficient case toward sheared media — an open problem in the mathematics of source identification. The physical claims are tested against direct numerical simulation; the mathematical results are derived in closed form and verified symbolically. The engineering consequences follow: attribution-invariant, code-independent prediction inputs that correct, at its root, an assumption on which interaction-noise prediction has rested for seventy years, with direct bearing on the computed noise of the wind turbines, rotors, propellers and turbomachinery of the energy transition. GENESIS aims not to improve an existing model, but to place the theory of how turbulence makes sound at a surface on new foundations.

Keywords: turbulence; rapid distortion; aeroacoustics; non-equilibrium turbulence; pressure–strain redistribution; acoustic analogy; source identification.



Graphical abstract. The upstream field Φ^∞ is carried by the rapid-distortion map H into the pre-impact field $\Phi^{LE} = H \Phi^\infty H^\dagger + N$; GENESIS asks whether the edge field is inherited or the residual N is generated in situ. The linearity spectrum $\gamma^2(\kappa)$ (right, schematic; measured curve in Fig. 2) gives the inherited fraction; the target scaling law is $N(\kappa)/N_0 = \mathcal{F}(\kappa r_{LE}; \vartheta, \chi)$.

Part I of the Scientific Proposal

1 State of the art: a founding assumption never tested

When a turbulent flow impinges on a solid body, a thin and strongly strained layer forms at the body's edge. What happens to the turbulence in that layer sets two quantities of first importance: the unsteady loads the body carries, and the sound it radiates. Turbulence–body interaction is the *dominant* broadband noise source of modern wind turbines, cooling and propulsion fans, contra-rotating propellers and rotorcraft, and of the distributed-propulsion and urban-air-mobility concepts on which low-carbon aviation depends; it also governs the unsteady gust loading that limits blade fatigue life. The underlying question, however, belongs to classical turbulence physics. How turbulence responds to strong mean strain has been a central problem of the field since Batchelor & Proudman [14], and the pressure–strain redistribution that governs that response remains the principal unclosed term of Reynolds-stress modelling [22, 23]. The bounded, inhomogeneous version of the problem — strain against a wall — recurs from wind-tunnel contractions to the stagnation region of every lifting surface, and it is the version no existing theory covers.

For seventy years this layer has been described by *rapid-distortion theory* (RDT). Because the mean flow accelerates through the region faster than the turbulent eddies can turn over, the fluctuations are assumed to obey a *linearised* equation: they are stretched and rotated by the mean strain, but not otherwise changed. In this picture the turbulence arriving at the surface is a *linear image* of the turbulence upstream. The assumption underpins the field's entire predictive framework. Amiet's theory — the standard method of interaction-noise prediction — multiplies an upstream-measured turbulence spectrum by a linear aeroacoustic transfer function; its modern refinements add finite thickness, real geometry and a *distortion correction*, but all insert the incident spectrum on the linear-inheritance premise [8, 28–30]. Recent rotor and wind-turbine distortion-modelling programmes likewise *model* the distortion in order to predict [31, 32]. Because the premise underlies the standard methods, it is rarely stated as a hypothesis — and it has not been tested at the level where it could fail.

It has never been tested directly, because the required measurement did not exist. To test it one must compare, *scale by scale*, the turbulence arriving at the surface against the turbulence upstream, and ask how much of the former is *linearly predictable* from the latter. That is a complex-valued, cross-spectral input–output measurement on an inhomogeneous straining field. It is beyond experiment — probes disturb the thin pre-impact layer and cannot return the full spectral tensor — and it was, until recently, beyond simulation: it needs both the resolution to capture the energy-containing scales through the distortion *and* a geometry that samples upstream and pre-impact states in vertex-for-vertex correspondence. Three separate bodies of prior work approach the question but stop short of it. (a) *Near-stagnation amplification*: grid turbulence is known to amplify as it is stretched by the diverging mean flow [17, 18], with a local production excess measured in canonical stagnation flows [21]. The question GENESIS poses is not whether fluctuations amplify — they do — but whether the pre-impact field remains a *linear image* of the upstream one, which is the operational premise the standard prediction chains rely on; these studies quantify *total* amplification and never separate an inherited part from a locally generated one. (b) *RDT with blocking*: the classical linear theory of turbulence near surfaces [15, 16] is, by construction, incapable of representing the non-linear terms that would generate turbulence — terms that become significant precisely under the strong straining considered here [20]. (c) *Non-equilibrium and stochastic closures*: recent Reynolds-stress models blend the RDT limit with return-to-isotropy [24, 25], and map-based stochastic models of the one-dimensional-turbulence family [19] resolve strained spectra at negligible cost — but both address *homogeneous, imposed* strain and the redistribution of *pre-existing* energy, not the bounded, inhomogeneous near-body field where turbulence may be *created*. Running underneath all of this is a second, older unease: the very notion of “the source” of aerodynamic sound is analogy-dependent and has been debated since Lighthill [6]; generalised acoustic analogies recast the ambiguity without removing it [10], Ffowcs Williams called the notion partly a “fiction” [7], and Tam showed acoustic analogies can identify the wrong source [11]. The specific objects at the heart of GENESIS — the *generated turbulence tensor*, its *universality*, and its *attribution-invariant* sound-producing content — are therefore unclaimed, and a direct, scale-resolved measurement of the generation is of standalone value to turbulence closure modelling, where the pre-impact pressure–strain term is exactly what closures must model but cannot easily measure. Posing that measurement's question precisely, and answering it, is where GENESIS begins.

2 The scientific question, the evidence, and the objectives

The question. *Is the sound-producing turbulence at a boundary inherited — a linear image of the incoming flow — or is it generated non-linearly in the straining near-field? And if it is generated: does the generation follow*

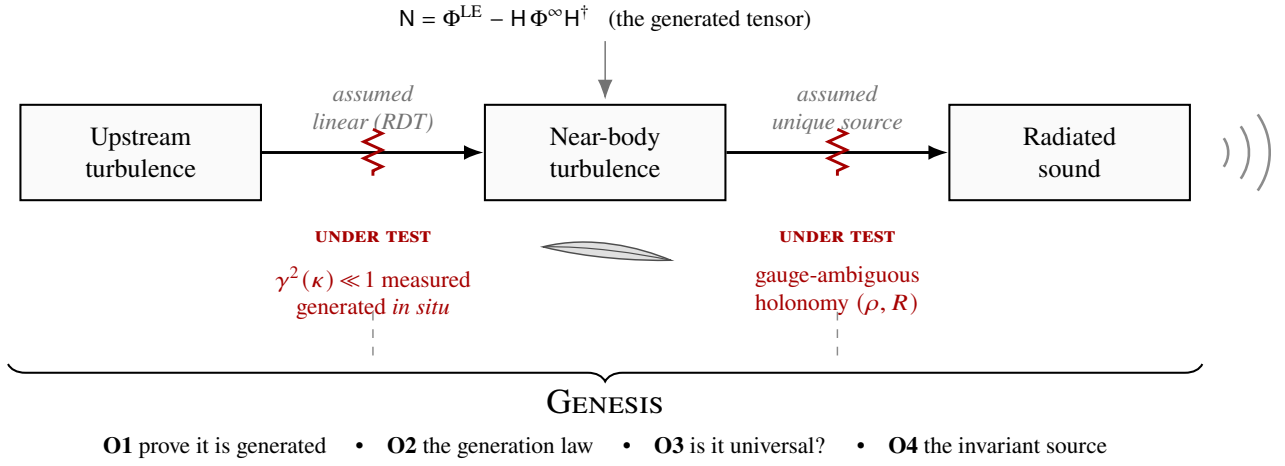


Figure 1: **The prediction chain, and the two links GENESIS places under direct test.** Prediction of turbulence–body interaction noise runs upstream turbulence $\rightarrow$ near-body turbulence $\rightarrow$ surface source $\rightarrow$ radiated sound. The field assumes the first link is *linear* (the turbulence is inherited) and the source is *unique*. Recent measurements and mathematics by the PI call both assumptions into question — the linearity spectrum γ^2 collapses [2], and source attribution carries an exact holonomy [1]. GENESIS’s four objectives place the chain on measured foundations.

a universal law of strained turbulence, and which part of the resulting field constitutes an invariant source of sound?

Preliminary evidence (the PI’s completed 2026 work — the springboard, not the project). The four results below, all published or under review, indicate that the assumption does not hold. (i) *The transformation is measurably non-linear.* From wall-modelled large-eddy simulation of grid turbulence convecting onto airfoils, sampled on nested vertex-matched toroidal surfaces enclosing the leading edge, the complex cross-spectral tensor between an upstream reference and the pre-impact state yields the input–output linearity spectrum $\gamma^2(\kappa)$ — the fraction of the pre-impact turbulence at each scale that is a linear image of the incoming field [2]. The result: $\gamma^2(\kappa)$ is bounded well below unity even at the largest scales ($\gamma^2 \lesssim 0.4$) and **collapses already at the energy-containing scales** ($\kappa \approx 0.3$) that set both the loading and the radiated noise — so most of the pre-impact tensor is generated in situ. A geometrically identical simulation with no body reproduces upstream isotropy and confirms this is physics, not a numerical or sampling artefact. (ii) *The anisotropy changes type in a way linear theory forbids* — the third anisotropy invariant crosses sign to a two-component state over the final few percent of the approach [2]. (iii) *The mechanism is a term absent from linear theory.* On the stagnation line the PI measured the pressure–strain redistribution directly: it drains the strain-amplified wall-normal stress into the spanwise and tangential components, is strongly spanwise-coherent, switches on exactly where the anisotropy changes type, and sets the spanwise-rich, large-scale character of the turbulence that ultimately radiates; across a **fourteen-fold sweep in leading-edge radius** it collapses on the nose radius, with a sharp bluntness threshold [2]. (iv) *The “source” of the sound is itself only defined up to convention — and that ambiguity admits a closed form:* re-attributing an aeroacoustic source between analogies is an exact geometric transport with a closed-form holonomy (a frequency dilation reaching 36% and a directivity rotation), so reusing one modelled spectrum across pipelines incurs a parameter-free inter-method bias that is pure convention, not physics [1].

Taken together, the four results point to a single conclusion: the linear picture of turbulence at a boundary does not hold at either end. The *input* to standard prediction is largely generated, not inherited; and the *output* object, “the source”, is gauge-ambiguous with a computable structure. But these are *ingredients, not the theory*. What they demand — a predictive law for the generated turbulence, a verdict on its universality, and an invariant definition of the source — does not yet exist. **Building it is GENESIS.**

Conceptual framework. Three quantities organise the whole programme (Fig. 1). The *linearity spectrum* $\gamma^2(\kappa)$ measures, scale by scale, how much of the near-body turbulence is a linear image of the upstream field — the fraction RDT assumes to be unity. The *generated tensor* $N = \Phi^{\text{LE}} - H \Phi^{\infty} H^\dagger$ is the residual — the turbulence created in the straining layer, which RDT assumes to be zero. And the *holonomy* of source re-attribution measures how much of “the source” is convention rather than physics. In the language of turbulence theory, γ^2 and N give the classical rapid-distortion problem its missing observables: they quantify, scale by scale, where the linear theory of strained turbulence ends and non-equilibrium generation begins. GENESIS is, in one sentence, the

GENESIS — THE EVIDENCE, THE QUESTION, THE PROGRAMME

(A1) NEAR-BODY TURBULENCE A LINEAR IMAGE OF THE UPSTREAM FLOW (RDT) · (A2) A UNIQUE ACOUSTIC SOURCE — BOTH UNDER DIRECT TEST

measured evidence (top row; 2026) shapes the four objectives (bottom row); colours as in the source papers
LINEARITY OF THE DISTORTION

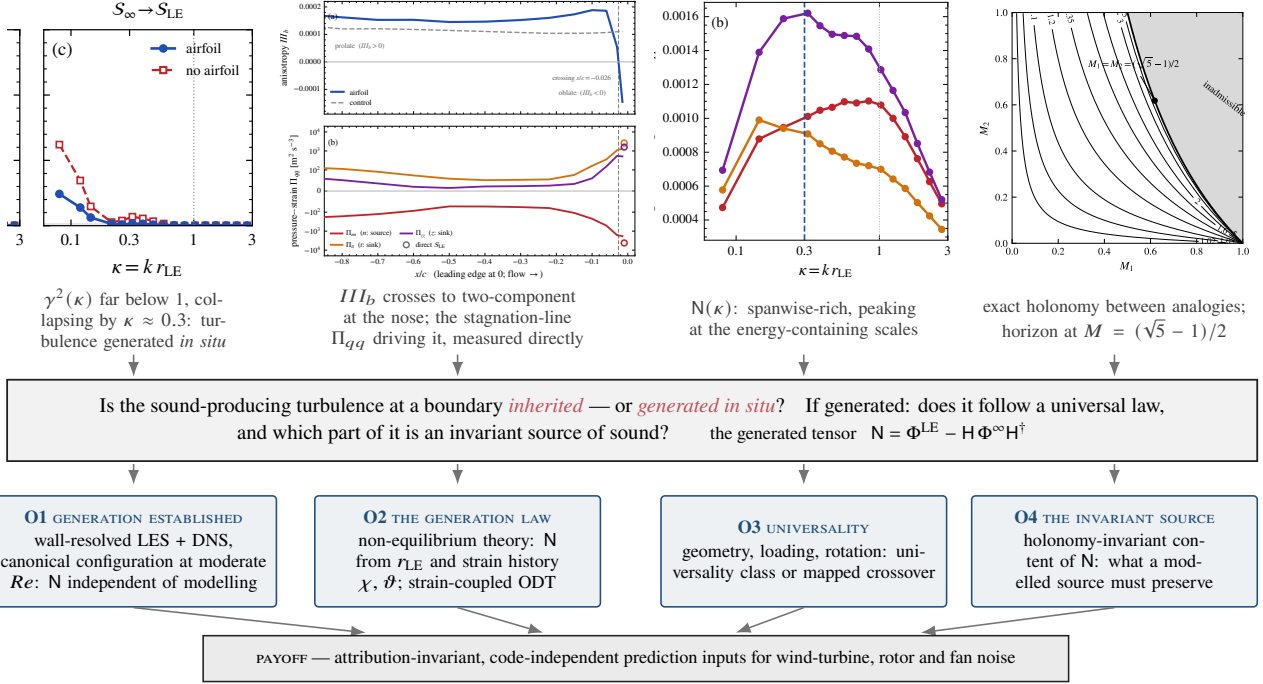


Figure 2: **The programme in one picture.** The four measured 2026 results [1, 2] (top row, with the measured behaviour sketched in each panel) place the two founding assumptions under direct test, pose the central question (middle band), and shape the four objectives (bottom row) — each objective standing directly beneath the result that motivates it. The objectives converge on the practical deliverable: attribution-invariant, code-independent prediction inputs.

programme that measures γ^2 and N , derives the law that governs N , tests that law for universality, and extracts from N the part that is holonomy-invariant and therefore physically radiates.

Central hypothesis. The sound-producing turbulence at a boundary is *generated*, *not inherited*, by a universal non-equilibrium process — a competition between mean-strain production and rapid pressure–strain redistribution in the bounded straining field, organised by a single geometric parameter (the nose-radius-scaled strain history) with the bluntness threshold as a genuine transition. The *attribution-invariant* part of the resulting source is exactly the part that sets the radiated far field, and is computable from the generation law.

Objectives. GENESIS is built backwards from the discovery it seeks, in four objectives — each beginning where the preliminary work ends — that answer, in turn, whether the effect is real, what law governs it, whether that law is universal, and which part of it is physically a source.

O1 The generation, established at the level of the governing equations. The wall-modelled measurement poses the question; **O1** answers it from first principles. Wall-resolved large-eddy simulation and direct numerical simulation of a *canonical* stagnation configuration at moderate Reynolds number — chosen precisely so that DNS is tractable and the physics isolated — determine $\gamma^2(\kappa)$ and $N = \Phi^{LE} - H \Phi^{\infty} H^\dagger$ free of subgrid and wall modelling, together with the full spectral budget of the pressure–strain term, a quantity no experiment can access. *Expected outcome:* the linearity boundary of rapid-distortion theory, established independently of turbulence modelling — a decisive early test of the programme’s premise.

O2 From mechanism to law. Derive the non-equilibrium theory of the bounded straining layer: a spectral evolution equation for $N(\kappa)$ in which mean-strain production competes with rapid pressure–strain redistribution, closed by the strain parameters $\chi = S k_t / \varepsilon$ and $\vartheta = \int S dt$ and reducing exactly to RDT in the rapid limit, with the PI’s strain-coupled one-dimensional-turbulence (ODT) model as the exploratory engine across the fourteen-fold radius sweep. *Expected outcome:* a predictive generation law with an exact linear limit, and the first directly measured stagnation-line pressure–strain benchmark for turbulence closure modelling.

O3 Universal? Test the law across a family of bodies (sharp→blunt airfoils, cylinders, plates), across mean loading — angle of attack, the field’s explicitly stated open case [30] — and from static to rotating blades,

where even the newest rotor-ingestion models still assume isotropic, undistorted inflow [32]; and determine whether the bluntness threshold is a genuine critical point or a smooth crossover. *Expected outcome:* the universality result — a new universality class of turbulence, or the mapped transition that replaces it — and the project's signature figure.

O4 An invariant, geometric theory of the source. Build the theory that decides which part of the generated field is physically a source. Using the PI's exact transport geometry of acoustic analogies [1], GENESIS defines the holonomy-invariant content of the generated tensor, proves that it — and only it — is what a *modelled* source must preserve to set the far field (measurable through the PI's diffraction-filtered surface sound-power source map [5]), and *extends the theory from the constant-coefficient case to sheared media*, where the transport becomes a Fourier-integral-operator groupoid and the invariant is conjecturally microlocal. *Expected outcome:* the first attribution-invariant definition of an aeroacoustic source — a foundational advance in the mathematics of source identification — and, from it, code-independent prediction with certified inter-method bias.

Why existing approaches cannot answer the question, and how GENESIS differs. The distinction is not incremental. The prevailing prediction frameworks — Amiet-type models and their distortion-corrected refinements [8, 28–30] — take the linear-inheritance premise as their starting point. Fitting a distortion *correction* to data is not a measurement of the *linearity of the transformation itself*: no prior study reports such a quantity. GENESIS instead subjects the premise to direct measurement and asks what physically replaces it where it does not hold. It therefore sits one conceptual level beneath those frameworks and would supply the corrected physics they consume, not a competing predictor. Linear RDT and its blocking extensions [15] cannot, by construction, represent the non-linear pressure–strain term that does the generating. The non-equilibrium closures closest in spirit [24, 25] treat *homogeneous, imposed* strain and redistribute pre-existing energy; GENESIS treats the *inhomogeneous, bounded, near-body* field and the *creation* of turbulence within it. And the mathematical theory of attribution-invariance exists only for free-space random sources [12]; GENESIS brings the minimal, physics-necessary slice — invariance under analogy re-gauging, tied to physically generated turbulence — rather than claiming to found that mathematics. GENESIS is thus unclaimed at every one of its four objectives simultaneously.

Broader significance and reach. The consequences extend well beyond noise. A directly measured stagnation-line pressure–strain term is exactly the quantity that Reynolds-stress and second-moment turbulence closures must model but can rarely measure, so O2's benchmark speaks to the wider turbulence-modelling community. A universal law for turbulence generated in a body's straining near-field (O3) would join grid decay and shear as a recognised route to anisotropic turbulence, with immediate bearing on unsteady gust loading and fatigue in wind-turbine and aero-engine blades. And an attribution-invariant notion of a source (O4) addresses a foundational question that has recurred across aeroacoustics since Lighthill, with practical consequence in certifiable, code-independent noise prediction — directly relevant to the low-noise wind-energy and distributed-propulsion technologies on which the energy transition depends. The project therefore turns a specific, sharp measurement into a broad conceptual and practical advance.

3 Overall approach and its ground-breaking nature

Answering the question takes four things: a way to *measure* the input–output map, scale by scale; a model economical enough to *explore* the physics across the whole parameter space; a criterion for which part of the measured field is truly a *source*; and an independent *check*. GENESIS has all four, each demonstrated at proof-of-concept in the PI's work, and directs them at one object — the generated tensor $N = \Phi^{\text{LE}} - H\Phi^{\infty}H^{\dagger}$ and its radiating trace: (a) scale-resolving simulation (wall-modelled, wall-resolved, a DNS anchor; lattice-Boltzmann on DLR HPC) *measures* N and γ^2 via the vertex-matched cross-spectral method of the leading-edge study; (b) the strain-coupled ODT model, exact in the RDT limit at $\sim 10^{-4}$ of LES cost, *models* N and supplies the closure to be tested [3]; (c) the exact transport geometry [1], with a diffraction-filtered surface sound-power quantity, makes the source content *invariant*; and (d) the verified vortex-particle estimator [4], which *computes* distortion from material vorticity conservation, cross-checks the linear/generated split independently. These four instruments have not previously existed in combination; it is their combination that makes the question answerable now.

The objectives are sequenced so that each stage is verified before the next depends on it, and each is anchored in a measured result (Fig. 2): O1 establishes the generation independently of turbulence modelling (the decisive early test); O2 turns the measured mechanism into a predictive law; O3 tests that law for universality — the central result — and O4 uses only the proven, uniform-flow slice of the transport geometry to identify the invariant source, with the more general variable-coefficient theory as upside rather than critical path. The programme

is *ground-breaking* in the strict sense the ERC asks for: not an improvement of an existing model, but the replacement, on measured evidence, of a founding assumption of an entire field — and the construction of the theory that succeeds it.

Why this is a five-year programme, not a paper. The preliminary measurements concern a single airfoil and a single simulation method; GENESIS turns that observation into a general theory, and the ambition rises year on year. **O1** does not merely repeat the measurement — it determines, with direct simulation, whether the effect survives at the true small scales, a question no wall-modelled study can answer. **O2** is not a model but a non-equilibrium theory of strongly strained turbulence, applicable where linear RDT and equilibrium closures cease to hold. **O3** asks whether boundary-induced generation constitutes a *new universality class* — testing, with measured exponents at the bluntness transition, a claim of the same kind as the canonical universality of grid or shear turbulence, not running a parametric sweep. **O4** pursues the most fundamental of the four: an attribution-invariant definition of a source, extended from the constant-coefficient case toward sheared media — an open problem in the mathematics of the field. And the phenomenon is not confined to leading edges: the same straining-boundary physics governs trailing edges, rotor and propeller tips, turbomachinery cascades, and wind-farm inflow, so the research line reaches well beyond the configuration in which it began. This is why the work needs five years, two doctoral researchers, and a PI who spans measurement, modelling and mathematics.

Risk, stated openly. The scientific risk is real, and it is structured to be informative: each central claim is falsifiable on a stated observable. If wall-resolved simulation shows γ^2 recovering toward unity at the energy-containing scales, the turbulence *is* inherited and the central hypothesis is refuted at the first objective (**O1**). If the normalised N does not collapse across geometries, universality fails — but a mapped two-branch transition would itself reshape how the field thinks about near-body turbulence (**O3**). If two flows with identical invariant content radiate differently, the invariance hypothesis fails (**O4**). Every one of these outcomes is scientifically decisive; none leaves the project without a result. The upside, if the hypothesis holds, is correspondingly large: a measured, non-linear, universal law where only a seventy-year-old linear assumption and brute-force simulation exist today.

Why now, and why this PI. The three enabling results are all recent, and all the PI's own: the first scale-resolved linearity measurement and the pressure–strain mechanism across a fourteen-fold geometry sweep [2]; the exact, machine-verified source geometry [1]; and the negligible-cost non-equilibrium instrument [3], with an independent estimator [4] and a proven fixed-and-rotating lattice-Boltzmann workflow on DLR HPC. The PI's formation as an applied mathematician (M.Sc. Applied Mathematics and Physics, MIPT; a self-directed mathematical-physics fellowship at Cambridge DAMTP) underwrites the theory, while the PI's DLR wind-energy setting and HPC access underwrite the campaigns. It is this combination of instruments that first allows the central question to be posed quantitatively.

What changes — theory first, then engineering. In one sentence: *if GENESIS succeeds, the field stops treating near-body turbulence as inherited and starts treating it as born there under its own law — and stops treating “the source” as a matter of convention and starts computing its invariant core.* Concretely, GENESIS delivers two foundational advances. In *turbulence physics*: the linear-inheritance assumption underlying decades of interaction-noise and gust-response modelling is retired, replaced by a non-equilibrium theory of strained turbulence and, potentially, a *new universality class* — boundary-induced genesis — with a directly measured pressure–strain benchmark for closure modelling. In the *mathematics of aeroacoustics*: the seventy-year-old ambiguity of “the source” is given exact structure and resolved into a holonomy-invariant object, the first attribution-invariant definition of an aeroacoustic source, reaching toward a microlocal theory on sheared media. *These theoretical and mathematical advances are the core of the project*; the engineering consequences follow from them — attribution-invariant, code-independent prediction that corrects, at the root, the noise models of the machines the energy transition depends on. What is *not* in dispute is the true radiated field — every exact analogy returns it identically; what GENESIS changes is the fundamental understanding of the *source*, and of the *input* on which prediction rests.

References

- [1] **Sharma, S.** (2026). A transport geometry of acoustic analogies: exact holonomy of source re-attribution and its observable consequences. *J. Fluid Mech.* (under review); arXiv:2608.12031.
- [2] **Sharma, S., Soni, M., Suryadi, A. & Herr, M.** (2026). Turbulence at an aerofoil leading edge: the linearity boundary and its generation mechanism. *J. Fluid Mech.* (under review); arXiv, submitted.
- [3] **Sharma, S.** (2026). Strain-coupled one-dimensional turbulence for rapid distortion. *J. Fluid Mech.* (under review); arXiv:2608.12048.

- [4] **Sharma, S.** & Herr, M. (2026). Distortion by construction: a verified vortex-particle estimator of the aeroacoustic transfer function. *Flow* (under review).
- [5] **Sharma, S.**, Soni, M. & Ayton, L. J. (2026). Where leading-edge noise is born: a significance-tested map of the acoustic source. *J. Fluid Mech.* (Rapids, under review).
- [6] Lighthill, M. J. (1952). On sound generated aerodynamically. I. General theory. *Proc. R. Soc. A* **211**, 564–587.
- [7] Ffowcs Williams, J. E. & Hawkins, D. L. (1969). Sound generation by turbulence and surfaces in arbitrary motion. *Phil. Trans. R. Soc. A* **264**, 321–342.
- [8] Amiet, R. K. (1975). Acoustic radiation from an airfoil in a turbulent stream. *J. Sound Vib.* **41**, 407–420.
- [9] Sears, W. R. (1941). Some aspects of non-stationary airfoil theory and its practical application. *J. Aeronaut. Sci.* **8**, 104–108.
- [10] Goldstein, M. E. (2003). A generalized acoustic analogy. *J. Fluid Mech.* **488**, 315–333.
- [11] Tam, C. K. W. (2002). Computational aeroacoustics examples showing the failure of the acoustic analogy theory to identify the correct noise sources. *J. Comput. Acoust.* **10**, 387–405.
- [12] Li, P., Wang, X., Wang, Y. & Zhao, Y. (2018). Inverse random source problems for time-harmonic acoustic and elastic waves. *Comm. Partial Differential Equations*.
- [13] Delfs, J. W. & Ruck, S. (2024). A quantity to identify turbulence related sound generation on surfaces. *J. Sound Vib.*
- [14] Batchelor, G. K. & Proudman, I. (1954). The effect of rapid distortion of a fluid in turbulent motion. *Q. J. Mech. Appl. Math.* **7**, 83–103.
- [15] Hunt, J. C. R. & Carruthers, D. J. (1990). Rapid distortion theory and the ‘problems’ of turbulence. *J. Fluid Mech.* **212**, 497–532.
- [16] Hunt, J. C. R. (1973). A theory of turbulent flow round two-dimensional bluff bodies. *J. Fluid Mech.* **61**, 625–706.
- [17] Bearman, P. W. (1972). Some measurements of the distortion of turbulence approaching a two-dimensional bluff body. *J. Fluid Mech.* **53**, 451–467.
- [18] Reynolds, A. J. & Tucker, H. J. (1975). The distortion of turbulence by general uniform irrotational strain. *J. Fluid Mech.* **68**, 673–693.
- [19] Kerstein, A. R. (1999). One-dimensional turbulence: model formulation and application to homogeneous turbulence, shear flows, and buoyant stratified flows. *J. Fluid Mech.* **392**, 277–334.
- [20] Kevlahan, N. K.-R. & Hunt, J. C. R. (1997). Nonlinear interactions in turbulence with strong irrotational straining. *J. Fluid Mech.* **337**, 333–364.
- [21] Sagaut, P. & Cambon, C. (2018). *Homogeneous Turbulence Dynamics*, 2nd edn. Springer.
- [22] Launder, B. E., Reece, G. J. & Rodi, W. (1975). Progress in the development of a Reynolds-stress turbulence closure. *J. Fluid Mech.* **68**, 537–566.
- [23] Speziale, C. G., Sarkar, S. & Gatski, T. B. (1991). Modelling the pressure–strain correlation of turbulence: an invariant dynamical systems approach. *J. Fluid Mech.* **227**, 245–272.
- [24] Brereton, G. J. (2024). “Return to equilibrium” anisotropy model for non-equilibrium Reynolds stress closures. *Phys. Fluids* **36**, 035121.
- [25] Clay, M. P., Buaria, D. & Yeung, P. K. (2016). A numerical study of turbulence under temporally evolving axisymmetric contraction and subsequent relaxation. *J. Fluid Mech.*
- [26] Lee, M. J. & Reynolds, W. C. (1985). Numerical experiments on the structure of homogeneous turbulence. Report TF-24, Stanford University.
- [27] Chen, J., Meneveau, C. & Katz, J. (2006). Scale interactions of turbulence subjected to a straining–relaxation–destraining cycle. *J. Fluid Mech.* **562**, 123–150.
- [28] de Santana, L. D. et al. (2016). A rapid distortion theory modified turbulence spectra for semi-analytical airfoil noise prediction. *J. Sound Vib.* **383**, 349–363.
- [29] Piccolo, A. et al. (2024). Turbulence distortion and leading-edge noise. *Phys. Fluids* **36**, 125183.
- [30] dos Santos, F. L. et al. (2023). Inflow turbulence distortion for airfoil leading-edge noise prediction for large turbulence length scales. *J. Acoust. Soc. Am.*
- [31] Yalçın, Ö. et al. (2026). Turbulence distortion matters in predicting inflow turbulence noise of future wind turbines. *Wind Energy Science*.
- [32] Raposo, H. & Azarpeyvand, M. (2024). Turbulence ingestion noise generation in rotating blades. *J. Fluid Mech.* **980**, A53.